

LuxeStay - Hospitality Website

Project Report

Overview

LuxeStay is a full-featured hotel booking platform with user authentication, room browsing, booking management, and a user profile dashboard. Built as a modern single-page web application deployed on Vercel with a serverless Python backend and Supabase for data persistence.

Pages

- Home - Hero section with search form, featured rooms, services, testimonials, stats
- Rooms - Room listing with filters (price, type, capacity)
- Room Details - Individual room view with gallery, amenities, booking form
- Booking - Checkout flow with personal details, payment summary
- About - Hotel story, stats, team, amenities, testimonials
- Contact - Contact form, info cards, FAQ accordion, map integration
- Login / Register / Forgot Password / Reset Password - Auth flow pages
- Dashboard - User overview with upcoming stay, recent stays, booking stats
- Profile - Full user profile with avatar, edit info, change password, booking & payment history

Tech Stack

Frontend

Technology

- HTML5
- Tailwind CSS (CDN)
- Google Fonts
- Material Symbols
- Vanilla JavaScript (ES6+)
- CSS Custom Properties

Purpose

- Page structure
- Utility-first styling
- Playfair Display + Inter typography
- Icon set
- Client-side logic
- Light / dark mode theming

Backend

Purpose

- Serverless API handler
- Serverless WSGI deployment
- Database persistence
- Authentication tokens
- Password hashing

Database (Supabase / PostgreSQL)

Purpose

- Accounts, profile fields, avatar
- Room inventory, pricing, amenities
- Reservations, dates, status, totals

Infrastructure

Purpose

Technology

- Python 3
- Vercel Python Runtime
- Supabase REST API
- JWT (HMAC-SHA256)
- SHA-256 + salt

Table

- users
- rooms
- bookings

Service

Supabase
Git

Hosting & serverless deployment
Managed PostgreSQL database
Version control

API Endpoints

Method	Path	Description
POST	/api/register	Create new user account
POST	/api/login	Authenticate and return JWT
GET	/api/profile	Fetch user profile
PUT	/api/profile/update	Update profile fields
POST	/api/profile/upload-image	Upload avatar (base64)
PUT	/api/change-password	Change user password
GET	/api/rooms	List all rooms
GET	/api/rooms/:id	Single room details
POST	/api/book	Create a booking
GET	/api/bookings	List user's bookings
GET	/api/bookings/:id	Single booking details
DELETE	/api/bookings/:id	Cancel a booking
POST	/api/contact	Submit contact form